

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

**A Web-Based Dairy Milk Billing and Management System for Automated
Purchase, Sales, and Supplier Record Management**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**A Web-Based Dairy Milk Billing and Management System for Automated Purchase, Sales, and Supplier Record Management**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Ajay Kumar**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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Date: 5th May 2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

In many dairy businesses, milk purchase records and billing are maintained manually, leading to calculation errors, inefficiency, and excessive paperwork. Pricing of milk depends on complex quality parameters such as fat content, CLR, and SNF, making manual management error-prone and costly. Existing software solutions are often too complicated for general use.

The objective of this project is to develop a web-based platform that automates milk purchase calculations and streamlines supplier management. The system enables real-time price calculation based on quantity and quality parameters. Developed using a full-stack JavaScript architecture (React.js, Node.js, and PostgreSQL), it provides automated billing, advance payment tracking, and secure record management. The expected outcome is improved accuracy, reduced paperwork, and streamlined business operations for dairy owners.

1. Introduction

1.1 Problem Background

1. Dairy businesses frequently maintain manual records for milk purchase and billing, which causes calculation errors and inefficiency.
2. Milk pricing is complex, relying on quantity, fat content, CLR, and SNF, leading to errors when done manually.
3. Manual systems often require external accountants, increasing operational costs for dairy owners.
4. Paper-based records result in duplication of work and significant time consumption.
5. There is a clear need for a user-friendly, real-time web-based system to automate these operations.

1.2 Objective of the Project

1. Develop a web-based Dairy Milk Billing and Management System.
2. Automate calculations for milk purchases based on quantity, fat, CLR, and SNF.
3. Simplify supplier management and tracking of advance payments.
4. Improve operational efficiency while reducing manual paperwork.

2. Methodology

2.1 Proposed Solution / System Overview

1. The proposed system utilizes a web-based platform designed to digitize various aspects of dairy management, including purchase, sales, and billing operations.
2. It is capable of performing real-time calculations for milk pricing by analysing specific quality parameters such as fat content and SNF.
3. The architecture incorporates several key components, including supplier management, purchase recording, and automated bill generation, to streamline daily activities.
4. The system leverages a secure database to ensure efficient record management and accurate financial tracking, providing dairy owners with a comprehensive and reliable digital solution.

3. Tools and Technologies

The system is developed using a full-stack JavaScript architecture.

1. **Frontend:** Built with React.js and Vite.
2. **Styling:** Tailwind CSS.
3. **Backend:** Node.js and Express.js.
4. **Database:** PostgreSQL.
5. **ORM:** Prisma ORM
6. **Communication:** RESTful APIs with secure authentication and integrated PDF generation.

4. Implementation

The "A Web-Based Dairy Milk Billing and Management System" was implemented using a modular, full-stack JavaScript approach to ensure scalability and maintainability.

4.1 Frontend Development (React.js)

The user interface was built using React.js with Vite for fast scaffolding and bundling. Tailwind CSS was employed for utility-first styling, ensuring a responsive and intuitive design. Key modules implemented on the frontend include:

- **Supplier Dashboard:** For managing supplier records, tracking advance payments, and viewing payment history.
- **Data Entry Forms:** To facilitate real-time input of milk purchase parameters (quantity, fat, CLR, SNF).
- **Billing Generation Interface:** Allowing users to filter records by date range and generate accurate, secure PDF bills on demand.

4.2 Backend Development (Node.js and Express.js)

The core business logic and API services are handled by a Node.js server utilizing the Express.js framework. The backend is responsible for:

- **Real-time Calculation Logic:** Implementing the complex algorithms to calculate milk price based on fat content, CLR, and SNF parameters.
- **RESTful API Services:** Defining secure endpoints for all CRUD (Create, Read, Update, Delete) operations related to suppliers, purchase records, and sales data. Secure authentication mechanisms are integrated to protect these endpoints.
- **PDF Generation:** Integrating a library to generate professional and secure PDF versions of the automated bills.

4.3 Database and ORM (PostgreSQL and Prisma)

PostgreSQL was chosen as the relational database for its reliability and robustness in handling financial data. Prisma ORM is used to interact with the database, providing a type-safe and efficient way to manage schema migrations and queries. The database stores critical information such as supplier profiles, daily milk purchase entries, price charts, and records of advance payments, ensuring accurate financial tracking.

5. Major Findings/Outcomes/Output/Results

The development and implementation of the Web-Based Dairy Milk Billing and Management System yielded several major outcomes and functional results that directly addressed the problems identified in manual dairy operations.

- **Automation of Complex Pricing:** The system successfully automated the complex calculation of milk purchases, accurately factoring in quantity, fat content, CLR, and SNF in real-time. This functionality minimized the calculation errors inherent in manual record-keeping.
- **Enhanced Operational Efficiency:** The transition from paper-based to a digital, web-based platform significantly streamlined purchase, sales, and billing operations. This increased overall efficiency and reduced the time previously consumed by duplicated work.
- **Secure Financial Tracking and Output:** The PostgreSQL database and integrated application logic provided a reliable system for secure record management, advance payment tracking, and accurate financial tracking. A key output is the on-demand generation of secure and accurate PDF bills for specified date ranges.
- **Reduced Operational Dependency and Cost:** By automating billing and record management, the system reduced the dairy owners' reliance on external accountants, thereby decreasing operational costs.
- **Comprehensive Management Modules:** The system delivered full-stack functionality, including user-friendly dashboards for supplier management, data entry forms for daily records (morning and evening), and flexible options for bulk rate updates and customer-specific pricing.

6. Conclusion and Future Scope

Conclusion

The Web-Based Dairy Milk Billing and Management System successfully achieved all its primary objectives by migrating complex manual dairy operations to a streamlined, digital platform. The successful implementation of a full-stack JavaScript architecture (React.js, Node.js, PostgreSQL) resulted in a robust system capable of automating real-time milk price calculations based on quality parameters (fat, CLR, and SNF). The major outcomes include a significant reduction in calculation errors, improved operational efficiency, secure record management for advance payments and financial tracking, and the ability to generate accurate, on-demand PDF bills. The project successfully delivered a user-friendly solution that significantly decreases the dairy owner's dependency on manual paperwork and external accounting personnel.

Future Scope

To further enhance the utility and reach of the system, the following areas are proposed for future development:

- **Mobile Application Integration:** Developing a native mobile application for suppliers and field staff using technologies like React Native to enable direct data entry and real-time viewing of payment summaries on the go.
- **Payment Gateway Integration:** Implementing secure payment gateway integration to facilitate direct digital payments to suppliers and collection from customers, further streamlining financial transactions.
- **Advanced Analytics and Reporting:** Expanding the dashboard functionalities to include advanced data visualization, trend analysis (e.g., seasonal changes in milk quality or yield), and forecasting tools to aid in strategic business decision-making.
- **Customer Relationship Management (CRM) Features:** Enhancing sales management modules with detailed customer profiling, personalized rate management, and automated communication features.

7. References

1. React.js Documentation. (n.d.). *Official Documentation for React*. Retrieved from <https://reactjs.org/docs>
2. Node.js Foundation. (2023). *Node.js API Reference*. Retrieved from <https://nodejs.org/en/docs/api/>
3. PostgreSQL Global Development Group. (n.d.). *PostgreSQL Documentation*. Retrieved from <https://www.postgresql.org/docs/>
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8. Appendices

This section includes supplementary material that supports the technical aspects of the project.

- **Appendix A: Source Code Snippets** - Key code blocks demonstrating the real-time calculation logic and API integration.
- **Appendix B: Database Schema Diagram** - A visual representation of the relationships between the Suppliers, Purchases, and Sales tables in the PostgreSQL database.
- **Appendix C: Sample PDF Bill Output** - An example of the automated, secure PDF bill generated by the system.